Monocular Depth Estimation as a Domain-Invariant Representation for Vision-Based Quadruped Parkour

Theo Lemay
Undergraduate Thesis
theo@example.edu

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Abstract

Vision-based quadruped parkour typically relies on onboard depth cameras, limiting deployment to robots with specialized sensors. We investigate whether a pre-trained *monocular depth estimator* can serve as a domain-invariant bridge, enabling sim-to-real transfer from only RGB without photorealistic rendering or domain randomization.

We present a two-phase pipeline: a privileged teacher trained via PPO with ground-truth terrain scans, distilled into a visual student via DAgger operating on depth images from Depth Anything V2. In simulation, we demonstrate that this pipeline achieves limited parkour capabilities under conditions characterized by a *Depth Signal Quality* metric. Domain invariance experiments show that DA2 provides strong robustness to color and brightness changes and generalizes across simulated environments.

1 Introduction

Legged robots that traverse irregular terrain—stairs, hurdles, gaps—must perceive geometry and mantain tight coordination between vision and control. Training locomotion policies in simulation with reinforcement learning (RL) and transferring them to real hardware has proven effective for agile skills [7, 9, 13], and recent work [2, 4, 6, 21] has extended this to parkour—climbing, jumping, and crawling—via teacher–student distillation. In these systems, a *teacher* trains with privileged terrain information (e.g. ground-truth heightmaps), then a *student* reproduces the teacher's behavior from onboard sensors.

A critical assumption, however, is access to an onboard depth sensor. Depth cameras (e.g. Intel RealSense) transfer well from simulation **??**, but not all platforms include them—the Unitree Go2, for instance, ships with only a front-facing RGB camera **??**.

Deploying RGB-trained policies on real robots requires bridging the *sim-to-real gap*. Three strategies have emerged: (1) **domain randomization** [17], which randomizes visual properties; (2) **generative augmentation** [20], which uses diffusion models for photorealistic training images; and (3) **canonical representations**, which project both domains into a shared space via a foundation model.

We pursue strategy (3) using Depth Anything V2 (DA2) [19], a pre-trained monocular depth estimator trained on millions of diverse real images. We propose that because DA2 maps arbitrary RGB to a depth space invariant to appearance, a student policy consuming depth should inherit that invariance without photorealistic rendering or explicit domain randomization.

Thesis statement. We hypothesize that pre-trained monocular depth estimation provides a domain-invariant intermediate representation for vision-based quadruped parkour, allowing a student trained on estimated depth in simulation to generalize across visual domains.

Contributions.

- A two-stage teacher–student pipeline from privileged scandot heightmaps to monocular depth for quadruped parkour.
- A demonstration that monocular depth enables basic parkour (stairs, hurdles, gaps) in simulation.
- Identification of depth signal quality—governed by model scale and terrain texture—as the key factor limiting performance.
- A domain invariance experiment testing DA2-based students under visual perturbations.

2 Related Work

RL for legged locomotion. Modern quadruped controllers are typically learned via reinforcement learning in simulation and transferred to real hardware [7, 13]. Chen et al. [3] introduced the privileged teacher–student paradigm, where a teacher with access to ground-truth state is distilled into a sensorimotor student. Lee et al. [9] applied this to locomotion, training a student from proprioceptive history alone and achieving zero-shot sim-to-real transfer on rough terrain. Miki et al. [10] extended this framework to perceptive locomotion, incorporating exteroceptive depth inputs.

Vision-based parkour. Agarwal et al. [1] introduced the scandot representation for egocentric depth-based locomotion, distilling a privileged teacher into a visual student. Extreme Parkour [4] and Robot Parkour Learning [21] build on this with CNN–GRU students that process depth images, demonstrating climbing, jumping, and crawling on real quadrupeds. SoloParkour [2] replaces distillation with off-policy RL warm-started from privileged experience, and ANYmal Parkour [6] uses a hierarchical skill-selection policy. All four systems assume access to a depth camera at deployment. Jenkins [8] is the closest work to ours, training an RGB parkour policy via domain randomization; we instead use monocular depth estimation as a canonical intermediate representation.

Sim-to-real for vision. Domain randomization [17] perturbs visual properties (textures, lighting, colors) so the policy learns invariance through exposure. LucidSim [20] replaces randomization with generative models that produce photorealistic training images. Xue et al. [18] combine photorealistic simulation with a teacher–student–bootstrap framework for humanoid loco-manipulation from RGB. All three approaches require significant rendering or generation overhead; we sidestep this by projecting RGB into a canonical depth space via a pre-trained estimator.

Monocular depth estimation. MiDaS [11] pioneered monocular depth estimation across diverse datasets. Depth Anything V2 (DA2) [19] achieves state-of-the-art zero-shot depth estimation using a vision transformer trained on 142M images. DA2 offers both relative and metric variants; we use the metric model, which predicts absolute depth in meters.

3 Mathematical Foundations

3.1 Problem Formulation

We model the parkour task as a Partially Observable Markov Decision Process $(\mathcal{S}, \mathcal{A}, \mathcal{O}, T, R, \gamma)$ with state space $\mathcal{S}$, action space $\mathcal{A}$, observation space $\mathcal{O}$, transition dynamics T, reward function R, and discount factor γ. The agent receives partial observation $o_t \in \mathcal{O}$ and selects action $a_t \in \mathcal{A} = \mathbb{R}^{12}$ (target joint position offsets). Observations decompose into three groups:

- **Proprioception** $o_{\text{prop}} \in \mathbb{R}^{53}$: base angular velocity, projected gravity, velocity and heading commands, joint positions, joint velocities, previous actions, and foot contacts.
- **Exteroception** o_{ext}: scandot heightmaps ($\mathbb{R}^{132}$) for the teacher, or depth/RGB images for the student.
- **Privileged** o_{priv}: friction and other physics parameters available only during teacher training.

3.2 PPO

The teacher policy π_θ is trained with Proximal Policy Optimization (PPO) [15], which maximizes a clipped surrogate objective:

$$L^{\text{CLIP}} = \mathbb{E}_t\Big[\min\Big(r_t \hat{A}_t,\ \text{clip}(r_t, 1{-}\epsilon, 1{+}\epsilon)\,\hat{A}_t\Big)\Big], \quad (1)$$

where $r_t = \pi_\theta(a_t|o_t)/\pi_{\theta_{\text{old}}}(a_t|o_t)$ is the probability ratio and $\hat{A}_t$ is the GAE [14] advantage estimate.

3.3 Reward Function

Table 1 lists the shaped reward terms.

Table 1: Reward function terms and scales.

Term	Scale	Description
tracking_goal_vel	3.0	Track goal velocity
tracking_goal_heading	1.5	Track heading
progress_along_course	2.0	Forward progress
waypoint_bonus	20.0	Sparse bonus
success_bonus	40.0	Course completion
termination	-50.0	Early termination
collision	-2.0	Body contact
feet_stumble	-1.0	Foot stumble
feet_edge	-2.0	Foot on edge
torques	$-2{\times}10^{-4}$	Torque penalty
dof_acc	$-2{\times}10^{-7}$	Joint accel.
action_rate	-0.01	Action smooth.
dof_pos_limits	-2.0	Joint limit
torque_limits	-0.2	Torque limit
lin_vel_z	-0.5	Vertical vel.
ang_vel_xy	-0.05	Roll/pitch rate
orientation	-0.5	Non-upright
flat_back	-3.0	Torso orient.

3.4 DAgger Distillation

The student policy is trained via Dataset Aggregation (DAgger) [12]. At each step the student observes o_{prop} and a depth image and produces a_{student}, but the teacher's action a_{teacher} is executed. The student minimizes:

$$\mathcal{L} = \underbrace{\|a_{\text{teacher}} - a_{\text{student}}\|_2}_{\text{action loss}} + \underbrace{\|y_{\text{oracle}} - \hat{y}\|_2}_{\text{yaw loss}}, \quad (2)$$

where $\hat{y} \in \mathbb{R}^2$ is the student's predicted heading and y_{oracle} is the ground-truth heading.

4 System Architecture

4.1 Overview

All experiments use the Genesis GPU simulator [5] with the legged_gym framework [13] and rsl_rl [16] on 4×

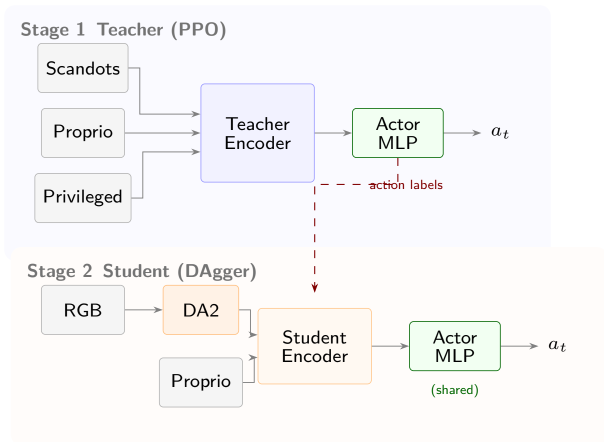

Figure 1: Two-stage training pipeline. The teacher uses privileged scandot observations; the student replaces these with either GT depth, DA2-inferred depth.

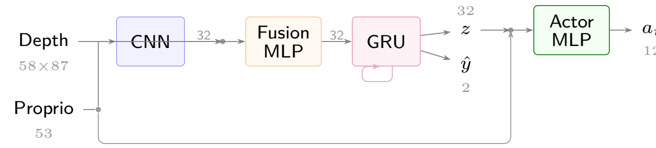

Figure 2: Student visual encoder. The depth CNN and proprioception are fused and processed by a GRU that produces the terrain latent and heading estimate used by the teacher's actor MLP.

[Terrain screenshots]
Stairs Hurdles Gaps

Figure 3: Simulated parkour terrains at increasing difficulty (left to right within each family).

NVIDIA L40S GPUs. The robot is a Unitree Go2 quadruped with 12 actuated joints. Actions $a_t \in \mathbb{R}^{12}$ are target joint positions, updated at 50Hz and converted to torques via a PD controller $\tau = K_p(a_t - q) - K_d\dot{q}$.

The system follows the two-stage privileged distillation pipeline described in Section 1: a teacher trains with ground-truth scandot heightmaps [1], then a student reproduces the teacher's behavior from depth or RGB images via DAgger (Figure 1).

4.2 Teacher

The teacher policy π_T maps proprioception, privileged physics parameters, and scandot heightmaps to joint targets. Three parallel MLPs encode each observation group into 32-dim latent vectors, which are concatenated with proprioception and fed through the actor MLP $\mathbb{R}^n \to \mathbb{R}^{12}$ to produce joint position offsets.

4.3 Student

The student replaces the scandot encoder with a visual encoder that must recover the terrain latent $z \in \mathbb{R}^{32}$ from a depth image (Figure 2). A CNN maps the depth image $d \in \mathbb{R}^{58\times 87}$ to a 32-dim feature, which is concatenated with proprioception and compressed by an MLP. A GRU integrates these fused features across timesteps, producing z and a heading estimate $\hat{y} \in \mathbb{R}^2$. The teacher's actor MLP is reused, taking $[o_{\text{prop}}; z]$ to output 12 joint targets. During DAgger distillation, the student minimizes the loss in Eq. 2 while the teacher's actions are executed in the environment.

4.4 Terrain and Curriculum

The simulator generates parkour terrains with three obstacle families (stairs, hurdles, gaps) at four difficulty levels per family (Figure 3) with gap widths from 0.24 m to 0.50 m. While these obstacles are simpler than those in prior work [2, 4, 21], our goal is to establish a baseline for evaluating the DA2 depth pipeline rather than to push the limits of agile locomotion. During teacher training, a curriculum advances robots to harder rows upon reaching a success threshold.

5 Experiments and Results

We evaluate the DA2 pipeline through three experiments: (1) establishing upper-bound baselines with ground-truth sensing, (2) closing the gap between estimated and ground-truth depth by varying model capacity and terrain texture, and (3) testing domain invariance under visual perturbations. We report *success rate* (fraction of episodes completing the full course) and *progress ratio* (mean fractional distance along the course before termination), both averaged over 256 episodes.

5.1 Baselines

Table 2 establishes upper bounds for the privileged teacher and the ground-truth depth sensing student (GT-depth). The privileged teacher achieves near-perfect success across all gap difficulties.

The GT-depth student, receiving simulator depth images, achieves ≥87% success on all rows, confirming the CNN–GRU architecture extracts sufficient geometry from a single depth viewpoint. A zero-depth ablation (all depth pixels set to zero) drops success to 8%, verifying the policy relies on depth rather than exploiting proprioceptive heuristics.

Figure 4 shows the teacher's training dynamics. Reward rises steeply as the policy masters flat terrain, then continues climbing as the adaptive curriculum exposes

Table 2: Baseline success rates on gap terrain (%).

	Row 0	Row 1	Row 2	Row 3
Scandots teacher	98.5	100.0	99.0	98.5
GT-depth student	94.5	94.5	89.1	87.5
Zero-depth ablation	8.0	2.0	0.0	0.0

progressively harder obstacles. By iteration 400 the curriculum has reached maximum difficulty and success saturates near 100%.

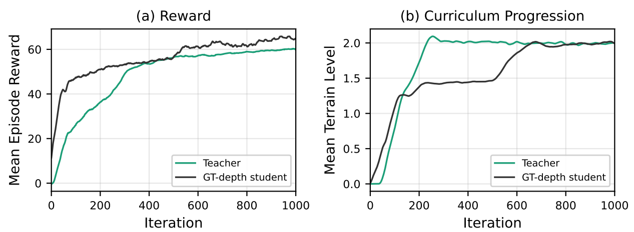

Figure 4: Teacher training curves. (a) Mean episode reward (solid) and success rate (dashed). (b) Mean terrain difficulty level assigned by the adaptive curriculum.

5.2 Parkour from Estimated Depth

This experiment tests whether parkour can be solved from estimated depth. We start with the simplest estimated-depth setup, then add the two changes that mattered most: a larger DA2 model and terrain texture.

The DA2-small baseline uses untextured terrain. It trains, but it does not converge to reliable gap crossing. DA2-base improves the easier rows, but the harder gaps still fail. Model scale helps, but it does not solve the core problem.

The remaining limitation is the visual input. On untextured terrain, the gap boundary provides weak monocular cues, so the depth estimate is ambiguous. Adding terrain texture makes the gap geometry visible. With DA2-base, this raises success to 80–93% across all rows and brings the estimated-depth policy within 7 pp of the GT-depth upper bound.

Table 3: Gap success (%) as model scale and terrain texture are added to the estimated-depth pipeline. GT-depth is shown as an upper bound.

Configuration	Row 0	Row 1	Row 2	Row 3
DA2-small	56.3	33.6	—	—
DA2-base	68.4	83.2	26.6	0.8
DA2-base + texture	**87.5**	**93.0**	**87.1**	**80.5**
GT-depth student	94.5	94.5	89.1	87.5

Table 3 summarizes this progression. Figure 5 shows the same pattern during training: the untextured models plateau early, while DA2-base + texture continues improving. Figure 6 shows that most of the gain comes from gaps; stairs and hurdles remain strong without ground texture because their vertical structure is already visible.

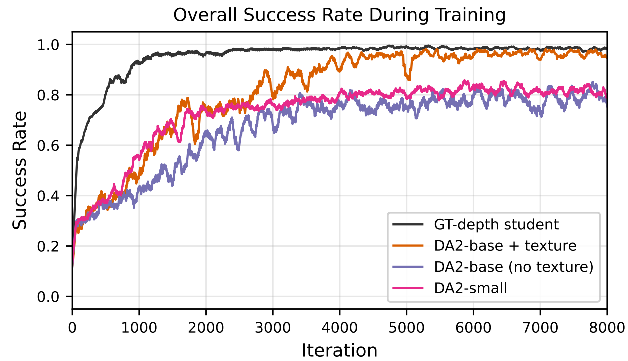

Figure 5: Overall success rate during student training. The untextured models plateau early, while DA2-base + texture approaches the GT-depth upper bound.

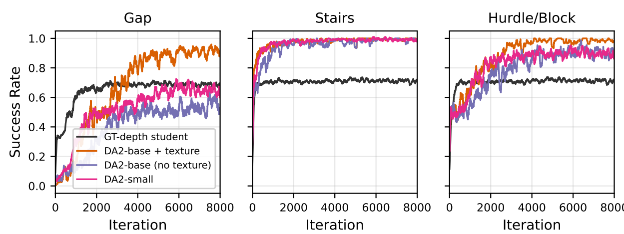

Figure 6: Per-obstacle success rate. Texture mainly improves gaps; stairs and hurdles are less sensitive because their geometry remains visible without ground texture.

5.3 Domain Invariance

The preceding results establish that DA2 enables strong parkour performance. We now test the core hypothesis: does the DA2 pipeline provide invariance to visual domain shifts?

Protocol. We apply perturbations at *test time only* to the trained DA2-base + texture student (Section 5.2). Perturbations span two categories:

- **Texture replacement**: the training checkerboard is replaced with structured alternatives (bricks, perlin noise) or featureless surfaces (no texture, solid colors, flat grey).
- **Color jitter**: global brightness scaling (±30%, ±60%) and per-channel color shifts (±15%, ±30%) applied to rendered RGB before DA2 inference.

We also test combined perturbations at mild and extreme severity. Each condition uses 128 episodes on gap Row 0.

Results. Table 4 shows a clean separation between two types of visual shift.

Table 4: Domain invariance: DA2-base + texture student on gap Row 0 under visual perturbations.

Perturbation	Succ.	Prog.	Δ
Baseline (checkerboard)	96.9%	.948	—
Texture replacement (structured)			
Bricks	89.8%	.908	−7.1
Noise	78.1%	.843	−18.8
Texture removal (featureless)			
No texture	18.0%	.437	−78.9
Solid red	23.4%	.471	−73.5
Solid blue	14.8%	.417	−82.1
Flat grey	18.0%	.456	−78.9
Color / brightness jitter			
Brightness ±30%	95.3%	.937	−1.6
Brightness ±60%	94.5%	.935	−2.4
Color ±15%	91.4%	.914	−5.5
Color ±30%	76.6%	.816	−20.3
Combined			
Mild (bricks+b30+c15)	82.0%	.862	−14.9
Extreme (noise+b60+c30)	63.3%	.747	−33.6

Analysis. DA2 is robust to color and brightness changes—the perturbations most relevant to sim-to-real transfer. Brightness jitter at ±60% causes only 2.4 pp drop; color jitter at ±15% causes 5.5 pp. DA2 also generalizes across structured textures (bricks: −7.1 pp).

Removing texture entirely causes catastrophic failure (−73 to −82 pp). This is not a domain invariance failure but an input quality requirement: monocular depth fundamentally needs visual features to infer geometry. Real-world surfaces (concrete, wood, soil) universally have natural texture, so this failure mode is an artifact of the simulator's ability to create unrealistically featureless geometry.

Combined perturbations compound but degrade gracefully: the mild condition (bricks + brightness 30% + color 15%) maintains 82.0% success, while the extreme condition drops to 63.3%.

6 Discussion and Conclusion

6.1 Summary

We presented a pipeline for vision-based quadruped parkour that uses monocular depth estimation as an intermediate representation. Our systematic experiments revealed that depth signal quality—governed by model scale and terrain texture—is the primary bottleneck for student performance, not network architecture. The best DA2-based student achieves 87% gap success, within 7 pp of the ground-truth depth upper bound.

The domain invariance experiment (Section 5.3) revealed a nuanced picture: the DA2 pipeline provides strong invariance to color and brightness perturbations (≤5.5 pp drop at moderate severity), which are the dominant sources of sim-to-real visual discrepancy. DA2 also generalizes across different structured textures (bricks: −7.1 pp). However, completely featureless surfaces cause catastrophic failure, reflecting a fundamental input requirement of monocular depth estimation rather than a domain invariance failure.

6.2 Implications

The DA2 pipeline provides effective domain invariance for the visual perturbations most relevant to sim-to-real transfer: lighting variation and color shifts. This suggests that a student policy trained on DA2 depth in a simple simulation (with any structured texture) could transfer to real-world environments without photorealistic rendering, provided the real surfaces have natural visual texture—which is nearly universally the case.

The key practical implication is a separation of concerns: the depth estimator handles domain invariance (absorbing appearance variation), while the student policy handles behavior (mapping depth to actions). This modularity means improvements to either component are independent: better depth models improve signal quality without retraining the policy; better training curricula improve behavior without changing the visual pipeline.

6.3 Limitations

- **No real-world deployment.** All experiments are in simulation. Real-world validation is needed to confirm domain invariance transfers beyond simulated visual shifts.
- **Computational cost.** DA2-base (100M params) runs at ∼40 ms/frame on an L40S GPU, which may be too slow for real-time deployment. Distillation into a smaller network or quantization could address this.
- **Texture requirement.** DA2 needs surface texture to produce useful depth estimates. Truly featureless surfaces (e.g., uniformly painted walls) may degrade performance. However, most real-world surfaces have natural texture.

6.4 Future Work

- **Real-world deployment** on the Unitree Go2 with onboard RGB camera.
- **Larger depth models**: DA2-large (335M params) or Video Depth Anything for temporal consistency.

- **End-to-end fine-tuning** of the depth estimator jointly with the student policy.
- **Multi-task evaluation** across diverse real-world environments (indoor, outdoor, varied lighting).

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